3. The protein part of the enzyme is known as 'apoenzyme'. The entire enzyme system consisting of the apoenzyme and coenzyme (or) prosthetic group is known as **holoenzyme**.

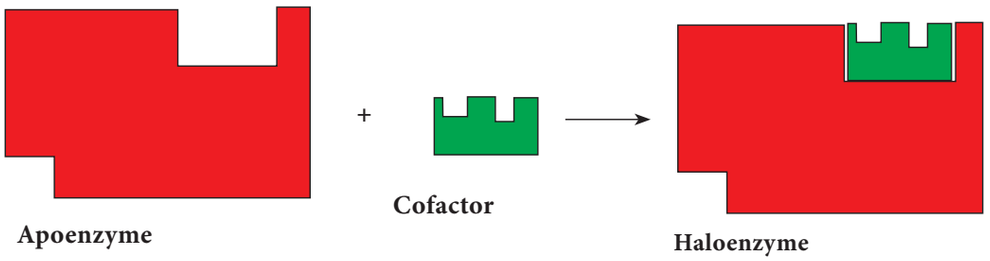

Figure 4.2 Apoenzymes and haloenzyme

4. Most of the coenzymes are linked to their apoenzymes by noncovalent forces. Example**:** The coenzyme ATP is attached to its apoenzyme hexokinase through weak non-bonding interactions.

5. Some of the coenzymes are tightly bound to their apoenzymes through covalent bonds. These are termed as prosthetic groups. Example: The prosthetic group biotin is attached to its apoenzyme carboxylase through a covalent bond.

6. The coenzymes undergo alterations during enzymatic reaction. Hence the coenzymes are regarded as second substrates (or) cosubstrates.

7. Many coenzymes are the derivatives of water soluble B-complex vitamins. **Examples:** Niacin.

8. The Coenzymes can also be organic substances other than vitamins. **Example :** ATP (adenosine triphosphate), CDP (Cytidine diphosphate)

9. Nucleotides and their derivatives can act as coenzymes.

   Example**:** NAD, FMN, FAD, coenzyme-A etc.

10. The specificity of an enzyme is mostly dependent on the apoenzyme and not on the coenzyme. For example, NAD^+ functions as coenzyme for several enzymes like alcohol dehydrogenase and lactate dehydrogenase.

11. Coenzymes functions as group transfer agents (Table 4.1).

Unit 4.indd 107 21-05-2018 15:10:06